Year 13 Maths - Radian measure

Similar questions - Sectors, iteration and small angles

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A sector OAB of a circle has radius 8 cm and angle θ radians. The area of the segment bounded by the chord AB and the arc AB is 10 cm^2.

(a) Show that $\theta = 0.3125 + \sin\theta$. **[3]**

(b) Explain, by considering the gradient of the right-hand side, why the iteration $\theta_{n+1} = 0.3125 + \sin\theta_n$ will converge to the root if the starting value is close enough to it. **[2]**

(c) Use the iteration with $\theta_1 = 1.5$ to find θ correct to 3 significant figures, showing the result of each step. **[3]**

(d) Use a change of sign to confirm that your answer to part **(c)** is correct to 3 significant figures. **[3]**

2

(a) For small θ (in radians), show that $3\cos\theta + 2\theta\sin\theta \approx 3 + \frac{1}{2}\theta^2$. **[3]**

(b) Hence find an approximate solution of $3\cos\theta + 2\theta\sin\theta = 3.02$. **[2]**

(c) Evaluate the left-hand side for your approximate value of θ and comment on the accuracy of the approximation. **[2]**

(d) A sector of a circle of radius 5 cm has perimeter 16 cm. Find the angle of the sector and the area of the sector. **[3]**

Question 1

(a) Segment area $= \frac{1}{2} \times 8^2(\theta - \sin\theta) = 32(\theta - \sin\theta) = 10$, so $\theta - \sin\theta = 0.3125$ and $\theta = 0.3125 + \sin\theta$

(b) The iteration function is $g(\theta) = 0.3125 + \sin\theta$ with $g'(\theta) = \cos\theta$. Near the root $\theta \approx 1.27$, $\cos\theta \approx 0.3$, and $|g'| < 1$, so the iteration converges (to a staircase pattern since $g' > 0$)

(c) $\theta_2 = 1.3100$, $\theta_3 = 1.2787$, $\theta_4 = 1.2701$, $\theta_5 = 1.2676$, $\theta_6 = 1.2669$, $\theta_7 = 1.2667$. So $\theta = 1.27$ to 3 s.f.

(d) Let $\mathrm{f}(\theta) = \theta - 0.3125 - \sin\theta$. $\mathrm{f}(1.2650) = -0.00111 < 0$ and $\mathrm{f}(1.275) = 0.00593 > 0$, so a root lies between 1.2650 and 1.275, confirming 1.27 to 3 s.f.

Question 2

(a) $\cos\theta \approx 1 - \frac{1}{2}\theta^2$ and $\sin\theta \approx \theta$: $3\left(1 - \frac{1}{2}\theta^2\right) + 2\theta^2 = 3 + \frac{1}{2}\theta^2$

(b) $3 + \frac{1}{2}\theta^2 = 3.02$, so $\theta^2 = 0.04$ and $\theta = 0.2$

(c) $3\cos 0.2 + 0.4\sin 0.2 = 3.0197$. This is very close to 3.02 (the error is about 0.0003), so the approximation is good for small θ

(d) Arc length $= 16 - 2 \times 5 = 6$ cm, so $\theta = \dfrac{6}{5} = 1.2$ rad. Area $= \frac{1}{2} \times 5^2 \times 1.2 = 15$ cm^2